HQL Lab Guide - Sorting, Pagination & Aggregates

Overview

This lab focuses on advanced HQL operations including sorting, pagination, filtering, and aggregate functions using the Product entity from Skill 2.

#SKILL - 3 ➔ Working with HQL - Sorting, Pagination & Aggregates

Prerequisites:

- **Basic knowledge of Hibernate**
- **Basic idea of HQL syntax**

In this lab, you will extend the Hibernate module created in **Skill 2** and perform advanced data retrieval using **HQL**, including **sorting**, **pagination**, **filtering**, and **aggregate functions**. These operations help in retrieving structured and meaningful data from the database using Hibernate.

Tasks:

1. Use the **Product entity** created in Skill 2.
2. Add **5–8 additional Product records** into the database.
3. Write HQL queries to retrieve all products sorted by price:
   a. Ascending order
   b. Descending order
4. Write an HQL query to sort products by quantity (highest first).
5. Implement pagination using HQL to display:
   a. First 3 products
   b. Next 3 products
6. Write HQL queries for aggregate operations:
   a. Count total number of products
   b. Count products where quantity > 0
   c. Count products grouped by description
   d. Find minimum and maximum price
7. Write an HQL query using **GROUP BY** to group products by description.
8. Write an HQL query using **WHERE** to filter products within a price range.
9. Write HQL queries using **LIKE** such as:
   a. Names **starting with** certain letters
   b. Names **ending with** certain letters
   c. Names containing a pattern **anywhere** (substring)
   d. Names with an **exact character length**
10. **Push the Hibernate project to GitHub**

Task 1: Use the Product Entity from Skill 2

Solution:

Ensure your `Product.java` entity is properly set up with the following structure:

```java
@Entity
@Table(name = "products")
public class Product {
    @Id
    @GeneratedValue(strategy = GenerationType.IDENTITY)
    private Long id;

    private String name;
    private Double price;
    private Integer quantity;
    private String description;

    // Constructors
    public Product() {}

    public Product(String name, Double price, Integer quantity, String descri
ption) {
        this.name = name;
        this.price = price;
        this.quantity = quantity;
        this.description = description;
    }

    // Getters and Setters
    public Long getId() { return id; }
    public void setId(Long id) { this.id = id; }

    public String getName() { return name; }
    public void setName(String name) { this.name = name; }

    public Double getPrice() { return price; }
    public void setPrice(Double price) { this.price = price; }

    public Integer getQuantity() { return quantity; }
    public void setQuantity(Integer quantity) { this.quantity = quantity; }

    public String getDescription() { return description; }
    public void setDescription(String description) { this.description = descr
iption; }

    @Override
    public String toString() {
        return "Product [id=" + id + ", name=" + name + ", price=" + price +
               ", quantity=" + quantity + ", description=" + description + "]
";
    }
}
```

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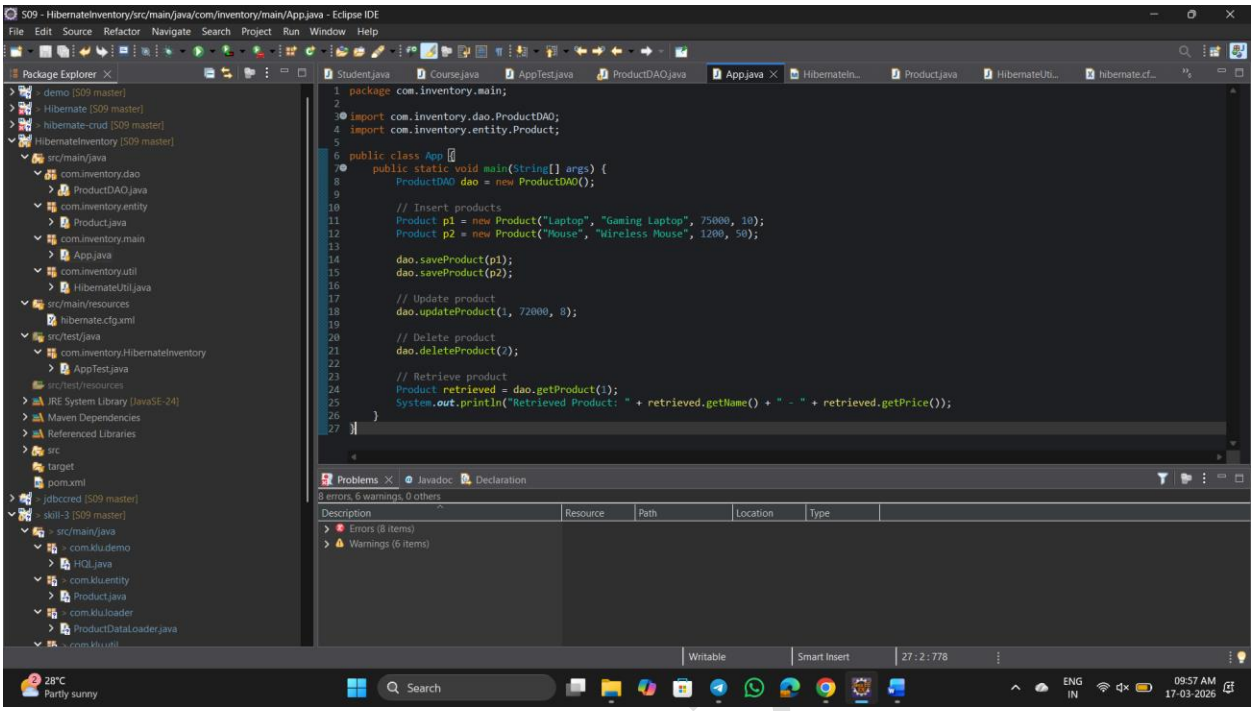

Task 2: Add 5-8 Additional Product Records

Solution:

Create a method to insert sample products:

```
public class ProductDataLoader {

    public static void loadSampleProducts(Session session) {
        Transaction transaction = session.beginTransaction();

        // Sample products
        session.save(new Product("Laptop", 899.99, 15, "Electronics"));
        session.save(new Product("Mouse", 25.50, 50, "Electronics"));
        session.save(new Product("Keyboard", 45.00, 30, "Electronics"));
        session.save(new Product("Monitor", 299.99, 20, "Electronics"));
        session.save(new Product("Desk Chair", 150.00, 0, "Furniture"));
        session.save(new Product("Desk Lamp", 35.75, 25, "Furniture"));
        session.save(new Product("Notebook", 5.99, 100, "Stationery"));
        session.save(new Product("Pen Set", 12.50, 75, "Stationery"));

        transaction.commit();
        System.out.println("Sample products added successfully!");
    }
}
```

Main method to execute:

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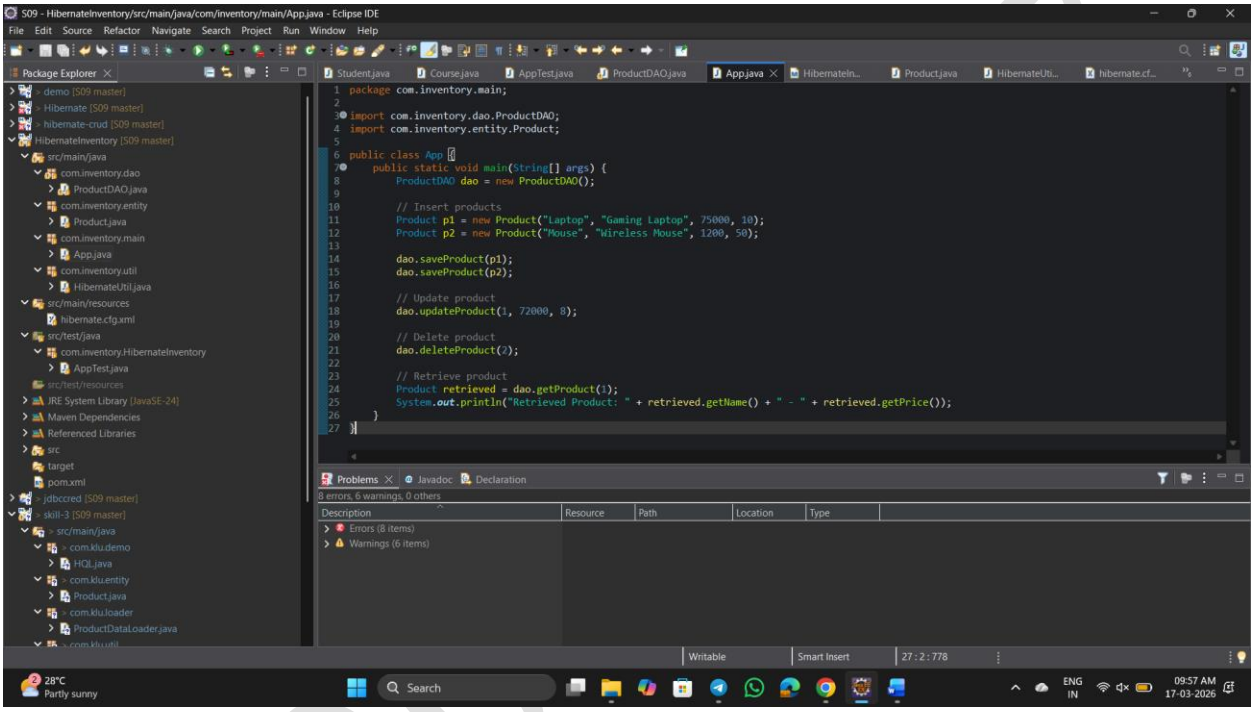
```java
public static void main(String[] args) {
    SessionFactory factory = HibernateUtil.getSessionFactory();
    Session session = factory.openSession();

    ProductDataLoader.loadSampleProducts(session);

    session.close();
    factory.close();
}
```



Task 3: Sort Products by Price

3a. Ascending Order

HQL Query:

```java
public static void sortProductsByPriceAscending(Session session) {
    String hql = "FROM Product p ORDER BY p.price ASC";
    Query<Product> query = session.createQuery(hql, Product.class);
    List<Product> products = query.list();

    System.out.println("\n=== Products Sorted by Price (Ascending) ===");
    for (Product product : products) {
        System.out.println(product);
    }
}
```

Expected Output:

```
Notebook - $5.99
Pen Set - $12.50
Mouse - $25.50
Desk Lamp - $35.75
Keyboard - $45.00
Desk Chair - $150.00
Monitor - $299.99
Laptop - $899.99
```

3b. Descending Order

HQL Query:

```java
public static void sortProductsByPriceDescending(Session session) {
    String hql = "FROM Product p ORDER BY p.price DESC";
    Query<Product> query = session.createQuery(hql, Product.class);
    List<Product> products = query.list();

    System.out.println("\n=== Products Sorted by Price (Descending) ===");
    for (Product product : products) {
        System.out.println(product);
    }
}
```

Expected Output:

```
Laptop - $899.99
Monitor - $299.99
Desk Chair - $150.00
Keyboard - $45.00
Desk Lamp - $35.75
Mouse - $25.50
Pen Set - $12.50
Notebook - $5.99
```

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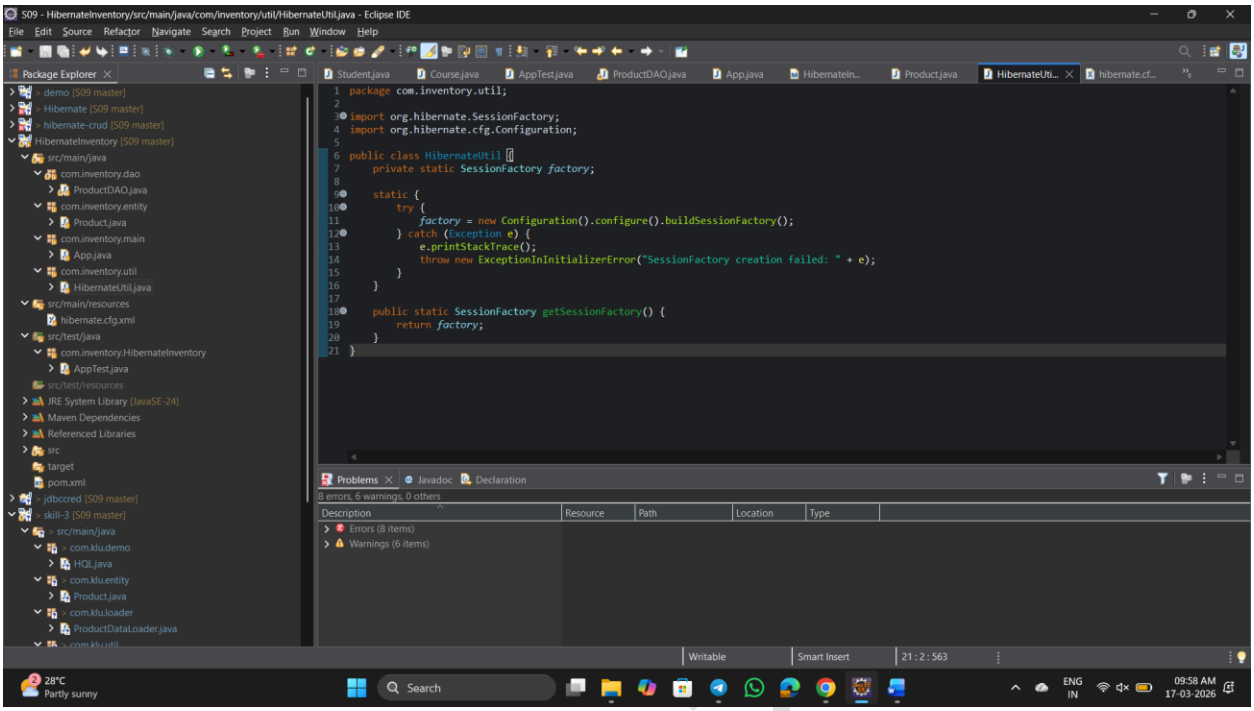

Task 4: Sort Products by Quantity (Highest First)

HQL Query:

```
public static void sortProductsByQuantityDescending(Session session) {
    String hql = "FROM Product p ORDER BY p.quantity DESC";
    Query<Product> query = session.createQuery(hql, Product.class);
    List<Product> products = query.list();

    System.out.println("\n=== Products Sorted by Quantity (Highest First) ===
");
    for (Product product : products) {
        System.out.println(product.getName() + " - Quantity: " + product.getQ
uantity());
    }
}
```

Expected Output:

```
Notebook - Quantity: 100
Pen Set - Quantity: 75
Mouse - Quantity: 50
Keyboard - Quantity: 30
Desk Lamp - Quantity: 25
Monitor - Quantity: 20
Laptop - Quantity: 15
Desk Chair - Quantity: 0
```

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Task 5: Implement Pagination

5a. First 3 Products

HQL Query:

```
public static void getFirstThreeProducts(Session session) {
    String hql = "FROM Product p";
    Query<Product> query = session.createQuery(hql, Product.class);
    query.setFirstResult(0);  // Start from index 0
    query.setMaxResults(3);   // Get 3 records

    List<Product> products = query.list();

    System.out.println("\n=== First 3 Products ===");
    for (Product product : products) {
        System.out.println(product);
    }
}
```

5b. Next 3 Products

HQL Query:

```
public static void getNextThreeProducts(Session session) {
    String hql = "FROM Product p";
    Query<Product> query = session.createQuery(hql, Product.class);
    query.setFirstResult(3);  // Start from index 3 (skip first 3)
    query.setMaxResults(3);   // Get 3 records

    List<Product> products = query.list();

    System.out.println("\n=== Next 3 Products ===");
    for (Product product : products) {
        System.out.println(product);
    }
}
```

Generic Pagination Method:

```
public static void getPaginatedProducts(Session session, int pageNumber, int
pageSize) {
    String hql = "FROM Product p";
    Query<Product> query = session.createQuery(hql, Product.class);
    query.setFirstResult((pageNumber - 1) * pageSize);
    query.setMaxResults(pageSize);

    List<Product> products = query.list();
```

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```
    System.out.println("\n=== Page " + pageNumber + " (Size: " + pageSize + "
) ===");
    for (Product product : products) {
        System.out.println(product);
    }
}
```

Task 6: Aggregate Operations

6a. Count Total Number of Products

HQL Query:

```
public static void countTotalProducts(Session session) {
    String hql = "SELECT COUNT(p) FROM Product p";
    Query<Long> query = session.createQuery(hql, Long.class);
    Long count = query.uniqueResult();

    System.out.println("\n=== Total Number of Products ===");
    System.out.println("Total Products: " + count);
}
```

Expected Output:

```
Total Products: 8
```

6b. Count Products Where Quantity > 0

HQL Query:

```
public static void countProductsInStock(Session session) {
    String hql = "SELECT COUNT(p) FROM Product p WHERE p.quantity > 0";
    Query<Long> query = session.createQuery(hql, Long.class);
    Long count = query.uniqueResult();

    System.out.println("\n=== Products in Stock (Quantity > 0) ===");
    System.out.println("Products in Stock: " + count);
}
```

Expected Output:

```
Products in Stock: 7
```

6c. Count Products Grouped by Description

HQL Query:

```
public static void countProductsByDescription(Session session) {
    String hql = "SELECT p.description, COUNT(p) FROM Product p GROUP BY p.de
scription";
```

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```java
    Query<Object[]> query = session.createQuery(hql, Object[].class);
    List<Object[]> results = query.list();

    System.out.println("\n=== Products Grouped by Description ===");
    for (Object[] result : results) {
        String description = (String) result[0];
        Long count = (Long) result[1];
        System.out.println(description + ": " + count + " products");
    }
}
```

Expected Output:

```
Electronics: 4 products
Furniture: 2 products
Stationery: 2 products
```

6d. Find Minimum and Maximum Price

HQL Query:

```java
public static void findMinMaxPrice(Session session) {
    String hql = "SELECT MIN(p.price), MAX(p.price) FROM Product p";
    Query<Object[]> query = session.createQuery(hql, Object[].class);
    Object[] result = query.uniqueResult();

    Double minPrice = (Double) result[0];
    Double maxPrice = (Double) result[1];

    System.out.println("\n=== Price Range ===");
    System.out.println("Minimum Price: $" + minPrice);
    System.out.println("Maximum Price: $" + maxPrice);
}
```

Expected Output:

```
Minimum Price: $5.99
Maximum Price: $899.99
```

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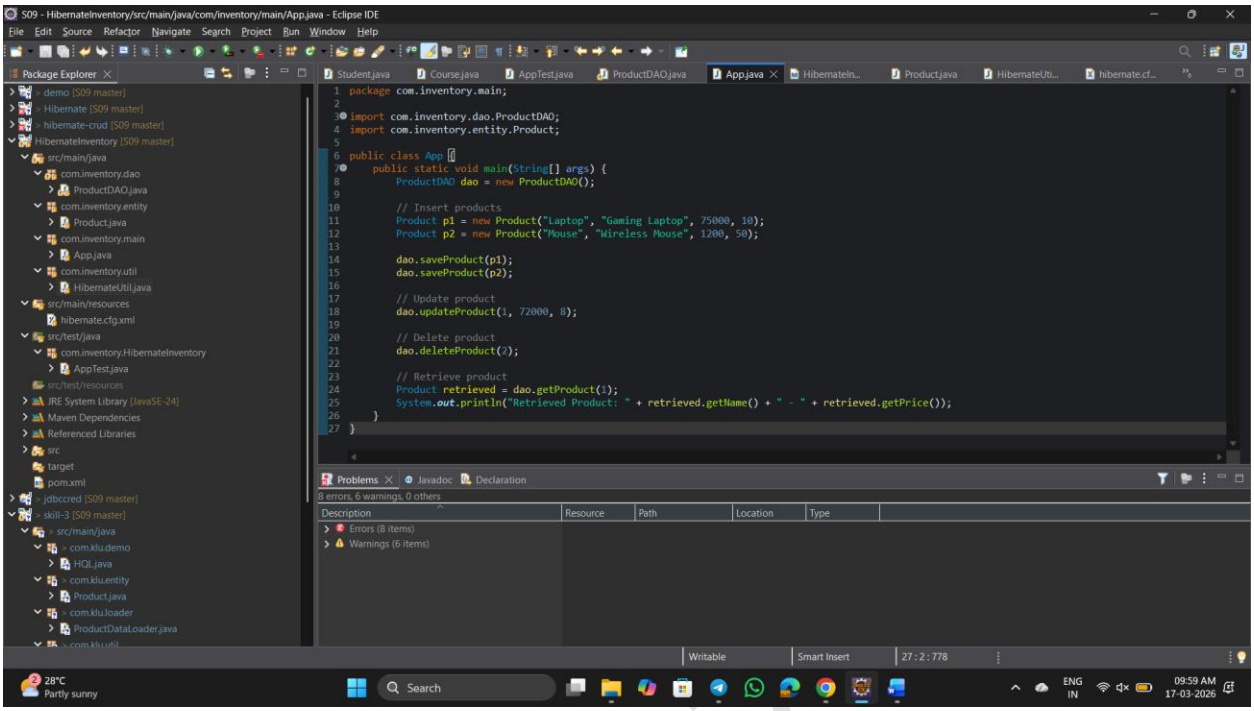

Task 7: GROUP BY to Group Products by Description

HQL Query:

```
public static void groupProductsByDescription(Session session) {
    String hql = "SELECT p.description, p.name, p.price FROM Product p ORDER
BY p.description";
    Query<Object[]> query = session.createQuery(hql, Object[].class);
    List<Object[]> results = query.list();

    System.out.println("\n=== Products Grouped by Description ===");
    String currentDescription = "";
    for (Object[] result : results) {
        String description = (String) result[0];
        String name = (String) result[1];
        Double price = (Double) result[2];

        if (!description.equals(currentDescription)) {
            System.out.println("\n" + description + ":");
            currentDescription = description;
        }
        System.out.println("  - " + name + " ($" + price + ")");
    }
}
```

Alternative with Aggregation:

```java
public static void groupProductsWithAggregation(Session session) {
    String hql = "SELECT p.description, COUNT(p), AVG(p.price), SUM(p.quantit
y) " +
                 "FROM Product p GROUP BY p.description";
    Query<Object[]> query = session.createQuery(hql, Object[].class);
    List<Object[]> results = query.list();

    System.out.println("\n=== Product Statistics by Category ===");
    for (Object[] result : results) {
        String description = (String) result[0];
        Long count = (Long) result[1];
        Double avgPrice = (Double) result[2];
        Long totalQuantity = (Long) result[3];

        System.out.println(description + ":");
        System.out.println("  Count: " + count);
        System.out.println("  Avg Price: $" + String.format("%.2f", avgPrice)
);
        System.out.println("  Total Quantity: " + totalQuantity);
    }
}
```

Task 8: Filter Products Within a Price Range

HQL Query:

```java
public static void filterProductsByPriceRange(Session session, double minPric
e, double maxPrice) {
    String hql = "FROM Product p WHERE p.price BETWEEN :minPrice AND :maxPric
e";
    Query<Product> query = session.createQuery(hql, Product.class);
    query.setParameter("minPrice", minPrice);
    query.setParameter("maxPrice", maxPrice);

    List<Product> products = query.list();

    System.out.println("\n=== Products Between $" + minPrice + " and $" + max
Price + " ===");
    for (Product product : products) {
        System.out.println(product.getName() + " - $" + product.getPrice());
    }
}
```

Usage Example:

```java
// Find products between $20 and $100
filterProductsByPriceRange(session, 20.0, 100.0);
```

Expected Output:

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```
Products Between $20.0 and $100.0:
Mouse - $25.50
Desk Lamp - $35.75
Keyboard - $45.00
```

Task 9: HQL Queries Using LIKE

9a. Names Starting with Certain Letters

HQL Query:

```java
public static void findProductsStartingWith(Session session, String prefix) {
    String hql = "FROM Product p WHERE p.name LIKE :pattern";
    Query<Product> query = session.createQuery(hql, Product.class);
    query.setParameter("pattern", prefix + "%");

    List<Product> products = query.list();

    System.out.println("\n=== Products Starting with '" + prefix + "' ===");
    for (Product product : products) {
        System.out.println(product.getName());
    }
}
```

Usage Examples:

```java
findProductsStartingWith(session, "D");  // Desk Chair, Desk Lamp
findProductsStartingWith(session, "M");  // Mouse, Monitor
```

9b. Names Ending with Certain Letters

HQL Query:

```java
public static void findProductsEndingWith(Session session, String suffix) {
    String hql = "FROM Product p WHERE p.name LIKE :pattern";
    Query<Product> query = session.createQuery(hql, Product.class);
    query.setParameter("pattern", "%" + suffix);

    List<Product> products = query.list();

    System.out.println("\n=== Products Ending with '" + suffix + "' ===");
    for (Product product : products) {
        System.out.println(product.getName());
    }
}
```

Usage Examples:

```java
findProductsEndingWith(session, "p");    // Laptop, Desk Lamp
findProductsEndingWith(session, "set");  // Pen Set
```

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9c. Names Containing a Pattern Anywhere (Substring)

HQL Query:

```
public static void findProductsContaining(Session session, String substring)
{
    String hql = "FROM Product p WHERE p.name LIKE :pattern";
    Query<Product> query = session.createQuery(hql, Product.class);
    query.setParameter("pattern", "%" + substring + "%");

    List<Product> products = query.list();

    System.out.println("\n=== Products Containing '" + substring + "' ===");
    for (Product product : products) {
        System.out.println(product.getName());
    }
}
```

Usage Examples:

```
findProductsContaining(session, "Desk"); // Desk Chair, Desk Lamp
findProductsContaining(session, "oo");   // Notebook
```

9d. Names with an Exact Character Length

HQL Query:

```
public static void findProductsByNameLength(Session session, int length) {
    String hql = "FROM Product p WHERE LENGTH(p.name) = :length";
    Query<Product> query = session.createQuery(hql, Product.class);
    query.setParameter("length", length);

    List<Product> products = query.list();

    System.out.println("\n=== Products with Name Length " + length + " ===");
    for (Product product : products) {
        System.out.println(product.getName() + " (Length: " + product.getName
().length() + ")");
    }
}
```

Alternative using LIKE with underscores:

```
public static void findProductsByExactLengthPattern(Session session, int leng
th) {
    StringBuilder pattern = new StringBuilder();
    for (int i = 0; i < length; i++) {
        pattern.append("_");
    }

    String hql = "FROM Product p WHERE p.name LIKE :pattern";
```

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```java
    Query<Product> query = session.createQuery(hql, Product.class);
    query.setParameter("pattern", pattern.toString());

    List<Product> products = query.list();

    System.out.println("\n=== Products with Exactly " + length + " Characters
===");
    for (Product product : products) {
        System.out.println(product.getName());
    }
}
```

Usage Example:

```java
findProductsByNameLength(session, 5);    // Mouse
findProductsByNameLength(session, 7);    // Monitor, Pen Set
```

Task 10: Push Hibernate Project to GitHub

Step-by-Step Guide:

Step 1: Create .gitignore File

Create a `.gitignore` file in your project root:

```
# Maven
target/
pom.xml.tag
pom.xml.releaseBackup
pom.xml.versionsBackup
pom.xml.next
release.properties

# Eclipse
.classpath
.project
.settings/

# IntelliJ IDEA
.idea/
*.iml
*.iws
*.ipr
out/

# NetBeans
nbproject/private/
build/
nbbuild/
```

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```
dist/
nbdist/

# Logs
*.log

# Database
*.db
*.sqlite

# OS
.DS_Store
Thumbs.db
```

Step 2: Initialize Git Repository

```
cd your-project-directory
git init
```

Step 3: Add Files to Git

```
git add .
git status
```

Step 4: Create Initial Commit

```
git commit -m "Initial commit: HQL Skill 3 - Sorting, Pagination & Aggregates
"
```

Step 5: Create Repository on GitHub

1. Go to GitHub.com
2. Click "New Repository"
3. Name it: `hibernate-hql-skill3`
4. Don't initialize with README (since you already have code)
5. Click "Create Repository"

Step 6: Connect Local to Remote

```
git remote add origin https://github.com/your-username/hibernate-hql-skill3.g
it
git branch -M main
git push -u origin main
```

Step 7: Create README.md

Create a `README.md` file:

```
# Hibernate HQL Skill 3 - Sorting, Pagination & Aggregates

## Overview
This project demonstrates advanced HQL operations including:
- Sorting (ASC/DESC)
- Pagination
```

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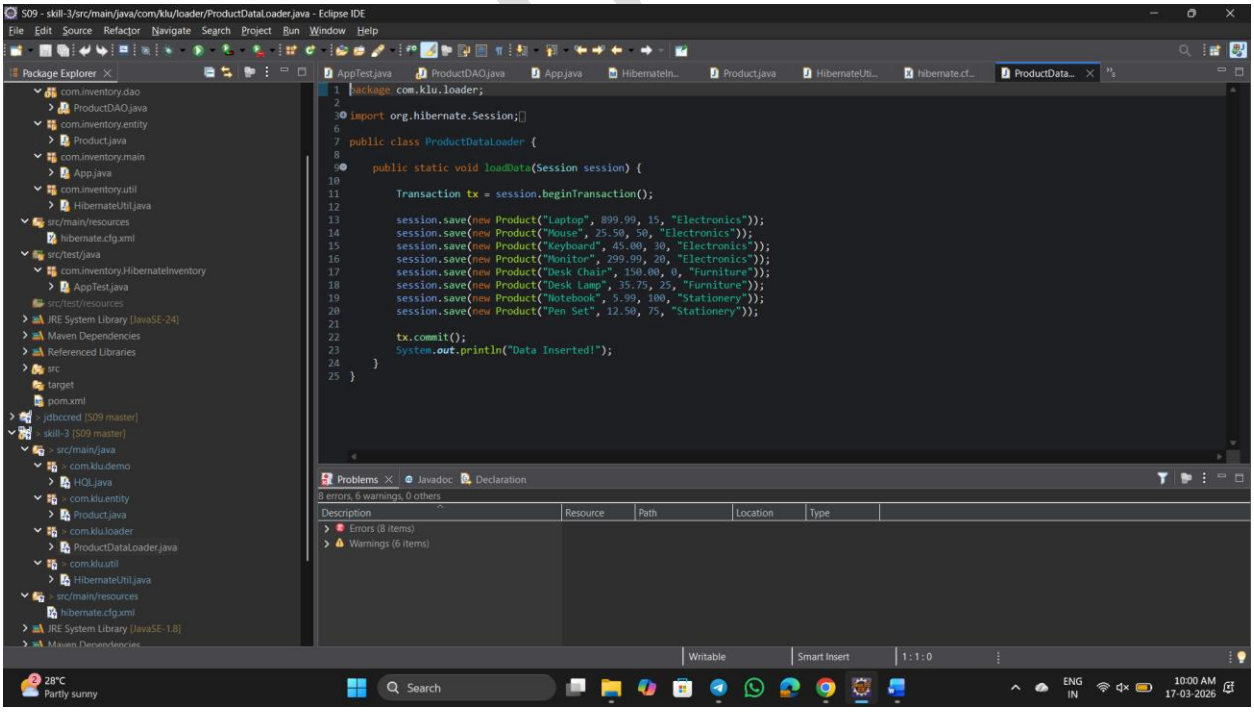
- Aggregate functions (COUNT, MIN, MAX, AVG, SUM)
- GROUP BY operations
- Filtering with WHERE
- Pattern matching with LIKE

Technologies
- Java 8+
- Hibernate 5.x
- MySQL/PostgreSQL
- Maven

Setup
1. Clone the repository
2. Update database credentials in `*hibernate.cfg.xml*`
3. Run `*ProductDataLoader*` to populate sample data
4. Execute queries from `*HQLDemo*` class

Tasks Completed
✅Product entity setup
✅Sample data insertion
✅Sorting by price and quantity
✅Pagination implementation
✅Aggregate operations
✅GROUP BY queries
✅Price range filtering
✅LIKE pattern matching



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```
## Author
[Your Name]
```

Step 8: Push README

```
git add README.md
git commit -m "Add README documentation"
git push
```

Project Structure on GitHub:

```
hibernate-hql-skill3/
├── src/
│   ├── main/
│   │   ├── java/
│   │   │   └── com/yourpackage/
│   │   │       ├── entity/
│   │   │       │   └── Product.java
│   │   │       ├── util/
│   │   │       │   └── HibernateUtil.java
│   │   │       ├── loader/
│   │   │       │   └── ProductDataLoader.java
│   │   │       └── demo/
│   │   │           └── HQLDemo.java
│   │   └── resources/
│   │       └── hibernate.cfg.xml
├── pom.xml
├── .gitignore
└── README.md
```

Complete Demo Class

Here's a complete demo class that runs all queries:

```java
public class HQLDemo {

    public static void main(String[] args) {
        SessionFactory factory = HibernateUtil.getSessionFactory();
        Session session = factory.openSession();

        try {
            // Task 2: Load sample data (run once)
            // ProductDataLoader.loadSampleProducts(session);

            // Task 3a: Sort by price ascending
            sortProductsByPriceAscending(session);

            // Task 3b: Sort by price descending
            sortProductsByPriceDescending(session);
```

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```java
            // Task 4: Sort by quantity
            sortProductsByQuantityDescending(session);

            // Task 5a: First 3 products
            getFirstThreeProducts(session);

            // Task 5b: Next 3 products
            getNextThreeProducts(session);

            // Task 6a: Count total products
            countTotalProducts(session);

            // Task 6b: Count products in stock
            countProductsInStock(session);

            // Task 6c: Count by description
            countProductsByDescription(session);

            // Task 6d: Min/Max price
            findMinMaxPrice(session);

            // Task 7: Group by description
            groupProductsByDescription(session);

            // Task 8: Price range filter
            filterProductsByPriceRange(session, 20.0, 100.0);

            // Task 9a: Names starting with 'D'
            findProductsStartingWith(session, "D");

            // Task 9b: Names ending with 'p'
            findProductsEndingWith(session, "p");

            // Task 9c: Names containing 'Desk'
            findProductsContaining(session, "Desk");

            // Task 9d: Names with exact length
            findProductsByNameLength(session, 5);

        } finally {
            session.close();
            factory.close();
        }
    }

    // ... (all the methods defined above)
}
```

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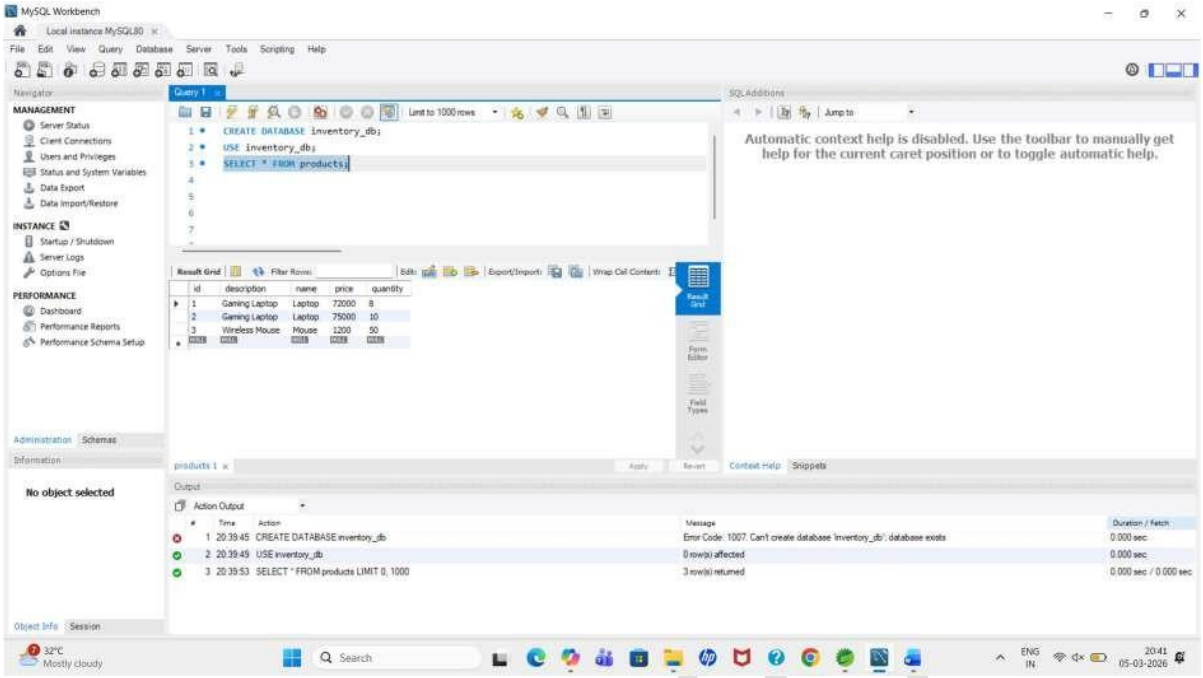

Testing Checklist

- ☐Product entity properly configured
- ☐8 sample products inserted
- ☐Sorting by price (ASC/DESC) works
- ☐Sorting by quantity works
- ☐Pagination returns correct results
- ☐All aggregate functions execute successfully
- ☐GROUP BY groups products correctly
- ☐Price range filtering works
- ☐All LIKE patterns match correctly
- ☐Project pushed to GitHub with proper .gitignore
- ☐README.md created and informative

Additional Resources

- Hibernate HQL Documentation
- HQL vs SQL Differences
- Hibernate Query Language Guide

GIT HUB OUTPUTS:

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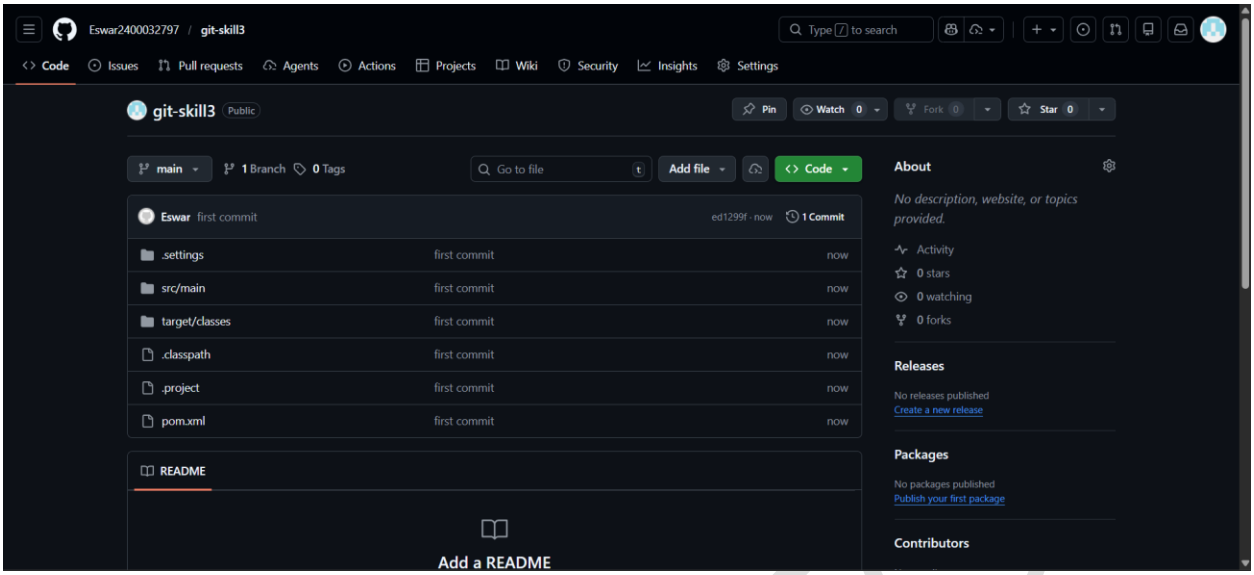

GIT REPO LINK:

https://github.com/Eswar2400032797/git-skill3.git

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